



EagleEye — Intelligent Edge AI Surveillance System

SparkUp / ORIC

assets/sparkup_logo.png

drop file → regenerate

Designer brief + embedded assets + drop-in slots ·

SparkUp Innovation Summit 2026 · 19 June 2026 ·

ITU ORIC

What this document is. A print-ready production pack for the A1 poster: the full design brief, the real images we already have (embedded below), clean drop-in slots for the images still needed, and a checklist of what's missing. Hand it to a designer or use it yourself in Canva/Figma. Style = **clean academic**.

Project	EagleEye — Intelligent Edge AI Surveillance System
Team	Murtaza Khalid (BSCE22004) · Huzaifa Khan (BSCE22025) · Haseeb Ahmed (BSCE22048)
Roles	Murtaza — Edge AI & Firmware · Huzaifa — Cloud & Mobile App · Haseeb — Audio Threat Detection
Supervisor	Dr. Rehan Hafiz (Professor; Director, ITU ORIC)
Co-Supervisor	Dr. Rehan Ahmed (Assistant Professor & Chairperson)
Affiliation	Dept. of Computer & Software Engineering, Faculty of Engineering — Information Technology University (ITU), Lahore
Deployed model	v7.16 — 96×96 RGB depthwise CNN, INT8 (on the ESP32-CAM)
Repo (QR)	github.com/muhammadAB123/fyp-eagle-eye

Headline metrics (deployed v7.16)

90.8%

detection accuracy (held-out test; 87.5% human recall, balanced)

<1 s

fully on-device inference (~872 ms, ESP32-CAM — no cloud, no GPU)

<3 s

detection → phone alert (end-to-end)

18

model versions engineered (v1.0 → v7.17)

~\$10

device vs \$200+ commercial · \$0/month cloud (free tiers)

100%

private — non-threat frames never leave the device

Judge-ready rigor line: "Selected from 18 trained variants — INT8 quantization, hard-negative mining, knowledge distillation, Optuna search, depthwise-separable architectures — converging on a custom from-scratch CNN holding >90% accuracy entirely on-device." (Peak across the line: 92.35%.)

Design spec (hard requirements)

Item	Spec
Canvas	A1 portrait, 594 × 841 mm , set at 300 dpi (7016 × 9933 px), 5 mm bleed, 15 mm safe margin
Export	PDF/X, CMYK , 300 dpi — confirm with the print shop before sending
Readability	Title readable at 3 m, headings at 1.5 m, body at 1 m. Title ≥ 90 pt · headings ≥ 42 pt · body ≥ 28 pt · captions ≥ 22 pt

Text budget	≤ 800 words total. Bullets / short phrases, not paragraphs. ~30–40% white space
Reading path	One path: top-left → down left column → center → right column → footer. Number sections 1–6

Palette & type

■ #1F4E79 primary blue · ■ #2196F3 accent teal · ■ #1A1A1A ink · ■ #555 caption · ■ #F4F6F8 panel · ■ #2E7D32 green · ■ #E53935 alert red. One clean sans-serif (Inter / Source Sans 3, Montserrat for headings). Bold headings, regular body; max 2 weights per size.

Layout (A1 portrait)

+-----+ HEADER BAND [ITU logo] EAGLEEEYE -- Intelligent Edge AI Surveillance one-line subtitle [SparkUp logo] team (roll#s) . supervisor . co-supervisor . ITU CSE +-----+	
(1) THE PROBLEM	(4) SYSTEM ARCHITECTURE *star* full cloud diagram, large
(2) OUR SOLUTION + hero device image	+-----+ (5) HOW IT WORKS (pipeline) Capture->AI->Alert->Phone
(3) KEY FEATURES (icon bullets)	+-----+ (6) RESULTS + TECH STACK metric tiles + logo row
+-----+ FOOTER: team . supervisors . ITU [QR demo][QR repo] Spark +-----+	

Section copy (use almost verbatim — already poster-length)

① The Problem

- **Cloud cams** stream your footage to a company's servers — privacy risk + monthly subscription.
- **DIY / LAN cams** work only on the same Wi-Fi — type the camera's IP, leave a PC running, risky port-forwarding.
- Most **record everything** — real events buried in hours of empty footage.

Goal: see real intrusions, from anywhere, privately, with no subscription.

② Our Solution

EagleEye runs a neural network **on the camera itself**. A low-cost **ESP32-CAM** decides on-device whether a human is present — images are analysed locally, not in the cloud. On an intruder it pushes a **photo alert to your phone**, and you can open an **on-demand live view** and **pan the camera** — from **any network, anywhere** (Wi-Fi or 4G/LTE). The camera **reaches out** to the cloud and stays connected, so there is **no static IP, no port-forwarding, no on-site PC**. End-to-end encrypted; runs entirely on **free tiers**.

③ Key Features

- **On-device human detection** — TinyML; images never leave the device to be analysed
- **Instant intruder alerts** — captured photo pushed to your phone, anywhere
- **On-demand remote live video** — stream only when you ask
- **App-controlled camera pan** — servo aim from the app
- **Arm / disarm + true online status** — real online/offline, not guesswork
- **Encrypted & private** — TLS everywhere, per-device credentials
- **~\$10 hardware · \$0/month** — entire cloud stack on free tiers

⑤ How It Works

Capture frame → On-device AI: human? → **yes** → Snap JPEG → Upload + Alert → Push to phone ; on request → Live video via cloud relay . **No** → keep watching.

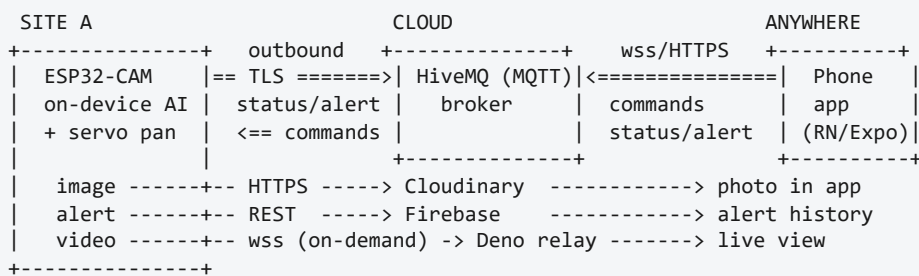
Only confirmed humans trigger an alert — no hours of empty footage.

⑥ Tech stack

ESP32-CAM · Edge Impulse (TinyML) · MQTT / HiveMQ Cloud · Cloudinary · Firebase · Deno Deploy relay · React Native + Expo · TLS / WebSocket.

Why novel: Edge AI on a microcontroller (privacy + zero inference cost) + an outbound-only architecture that works on any network — the pattern commercial IoT (AWS IoT, Ring/Nest) uses, rebuilt at \$0/month.

④ System Architecture (the centerpiece — redraw clean & flat)



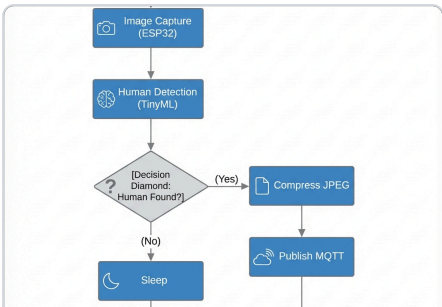
Every arrow points OUT from the device. Put a small lock icon on each link (end-to-end TLS).

Asset board — what we have (embedded) & what we need (slots)

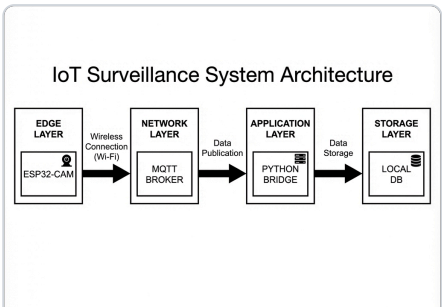
Embedded — already on disk



Hero device (render) — have
Stylised 3D render. Use as hero *only if* no real photo.
assets/hardware_setup.png



Detection pipeline — have
Clean, usable for §5 as-is.
assets/local_flowchart.png



Architecture (OLD/LAN) — redraw
Shows old bridge+localDB. Redraw to cloud (§4).
assets/system_architecture.png



"Intruder detected" HUD — inset only
Cyberpunk; too sci-fi for clean style.
Small inset max.
assets/alert_interface.png

Real captures: backend/captures/ has 50+ genuine ESP32-CAM detection JPEGs, but most are flash-overexposed and rotated — not poster-grade. Take one clean fresh capture for the "real detection" slot below.

Needed — drop the file into docs/assets/ with the exact name, then ask me to regenerate

Real prototype photo
assets/prototype_photo.jpg
your actual built ESP32-CAM unit (hero)

App screenshot
assets/app_screenshot.png
dashboard or live-view screen

Real detection capture
assets/detection_sample.jpg
one clean intruder photo from the system



INFORMATION
TECHNOLOGY
UNIVERSITY

ITU logo — have
Embedded in header.

SparkUp / ORIC logo
assets/sparkup_logo.png
event branding

Demo-video QR
paste the video URL – I'll
generate the QR
e.g. YouTube/Drive link

Still missing before print — checklist

- ☐ **Real prototype photo** of the built ESP32-CAM unit → save as `docs/assets/prototype_photo.jpg`
- ☐ **App screenshot** (dashboard + live view) → `docs/assets/app_screenshot.png`
- ☐ **One clean detection capture** (not overexposed) → `docs/assets/detection_sample.jpg`
- ☒ **ITU logo** — added `docs/assets/itu_logo.png`
- ☐ **SparkUp / ORIC logo** → `docs/assets/sparkup_logo.png`
- ☐ **Demo-video URL** (for the QR) → just paste it to me
- ☐ **Confirm** the GitHub repo is public for the repo QR → `github.com/muhammadAB123/fyp-eagle-eye`

Drop the files in and tell me "regenerate" — they'll appear embedded in place of the dashed slots, and I'll add the QR codes.

Print & display checklist (event day)

- Export **PDF/X · CMYK · 300 dpi · A1 · 5 mm bleed** — confirm with the print shop first.
- Proof at 100% on screen; print one **A4 test** to catch low-res images / clipped text.
- Get it **A1 printed + framed** (assessment requirement, 10% of FYP grade).
- Bring your **own laptop + LCD/monitor + mounting** for the live demo. **Extension cords provided by ORIC.**
- Flash the unit with the **v7.16** model so the demo matches the poster's 90.8% / ~872 ms numbers.